GRAVITYSORT · PROJECT PLAN

GravitySort project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline): workstreams, resources, stage gates P0 to P6 (P4 to P6 on hold), next steps, decisions requested and the reference list

GravitySort project plan

GravitySort is at stage P3 (design outputs, TRL 3): the constructable design, its D3 model of every concept option, drawings, priced BOM, build plan and pipeline documents exist on paper. The next gate, P4 (verification, TRL 4), is on hold under the portfolio cap; before it Amish is asked to confirm the concept (option A recommended, GVS-DEC-001 O1) and to decide the containment band (O2). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and STEM	Design lead	GVS-STM-001, GVS-OPT-001, GVS-DDR-004	Done at P3; STEM check open: nine rows need tests or datasheets, two need Amish's decisions (O2, O3)
Requirements and traceability	Design lead	GVS-URS-001, GVS-PRS-001, GVS-RTM-001	Done at P3; to revise after co-design
Calculations	Design lead	GVS-CAL-001 v0.9	Done at P3
CAD and drawings	Design lead	`cad/src/model.py` (D3, options A to C), STEP, GVS-DWG-001 to 121	Done at P3
Build plan and BOM	Design lead	GVS-BLD-001, `bom/bom.csv`	Done at P3
Regulatory, risk and market	Project lead	GVS-REG-001, GVS-RSK-001, GVS-MKT-001	First issue; items to assess with the partner
Partner and co-design	Project lead (Amish)	Agreement with the first candidate cooperative; co-design sessions	Not started (outreach is Amish's action)
Renders and media	Coordinator	Options picture, product renders, cards (Blender on Amish's Mac)	Scenes ready; renders pending
Prototype build and lab tests (P4)	Workshop partner and design lead	Prototype, test plans and reports	On hold (TRL cap 3)
Field trial (P5)	Partner organization	Trial with the cooperative	On hold

Resources needed

Table 2. Resources (estimates, to assess with the partner before P4).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts	One machine to the BOM	n/a	594 (BOM estimate) plus 335 if the MotionCore option is built
Workshop time (welder, fitter)	Frame, guards, table, assembly	To assess with the workshop	To assess
PU casting trial	Liner cast in the printed mold	To assess	To assess
Spiked or characterized test ore and assays	Recovery test of PR-004	To assess	To assess
Partner co-design sessions	Validated user requirements	To assess	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (2026-10-25, retrofitted under kit 1.8)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Met (2026-10-25)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed where a calculation can close it	Met on paper; concept selection awaiting Amish (GVS-DDR-004)
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability complete	Met (TRL 3 since 2026-09-25; D3 model of every option 2026-10-25)
P4 Verification	Prototype, lab test plans and reports (recovery, table ratio, pedal power, containment impact test)	On hold (TRL cap 3)
P5 Validation	Field trial with the partner, human factors	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. **Decision O1 (concept selection, GVS-DDR-004):** confirm option A, the current design, as the concept carried forward (recommended), or choose B or C.
2. **Decision O2 (containment):** keep the 6 mm tub with an impact test before any spin test (recommended), or fit a steel band now, which takes the mass over PR-011 (GVS-DEC-001).
3. **Decisions O3 to O5:** tank sway near the table frequency (raise the brace node and take a saving, recommended), spindle supports (tap test at TRL 4), fluidization flow (sweep it in the TRL 4 test).
4. Amish's outreach to the first candidate partner (decided 2026-10-02), so that co-design can test the user requirements.

5. Coordinator renders `media/options.png` and the product renders on Amish's Mac.
6. When the TRL cap lifts: the P4 tests in Table 3, starting with the containment impact test and the recovery test.

References

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